

# Comprehensive End-to-End System Documentation & Comparative Study

## Auditing Agency Performance Monitoring, Classification, Scoring, and Air-Gapped Ranking Module

**Project:** AI-Based Solution for Analysing, Benchmarking, and Quality Monitoring of Cybersecurity Audit Reports and Performance Monitoring of Auditing Organisations

**Submission Package Identifier:** PerformanceMonitoring\_Saket\_20260824

**Date of Submission:** 2 September 2026

### Executive Summary & Complete Deliverables Architecture (D1–D13)

This technical document serves as the authoritative, unified design and evaluation reference for the clean-room implementation of the **Auditing Agency Performance Monitoring, Scoring, and Ranking Module**, fully addressing all thirteen deliverables (**D1 through D13**).

The system evaluates the performance of empanelled cybersecurity auditing organisations on a continuous basis, computes six mandatory performance features from raw operational telemetry, trains and compares four renewal outcome classification models, synthesizes a continuous hybrid performance score (combining multi-attribute features and model-predicted renewal probabilities), executes deterministic ranking with multi-tier tie-breaking, detects decision boundary uncertainty, and provides immutable versioning and end-to-end provenance in a strictly air-gapped offline environment.

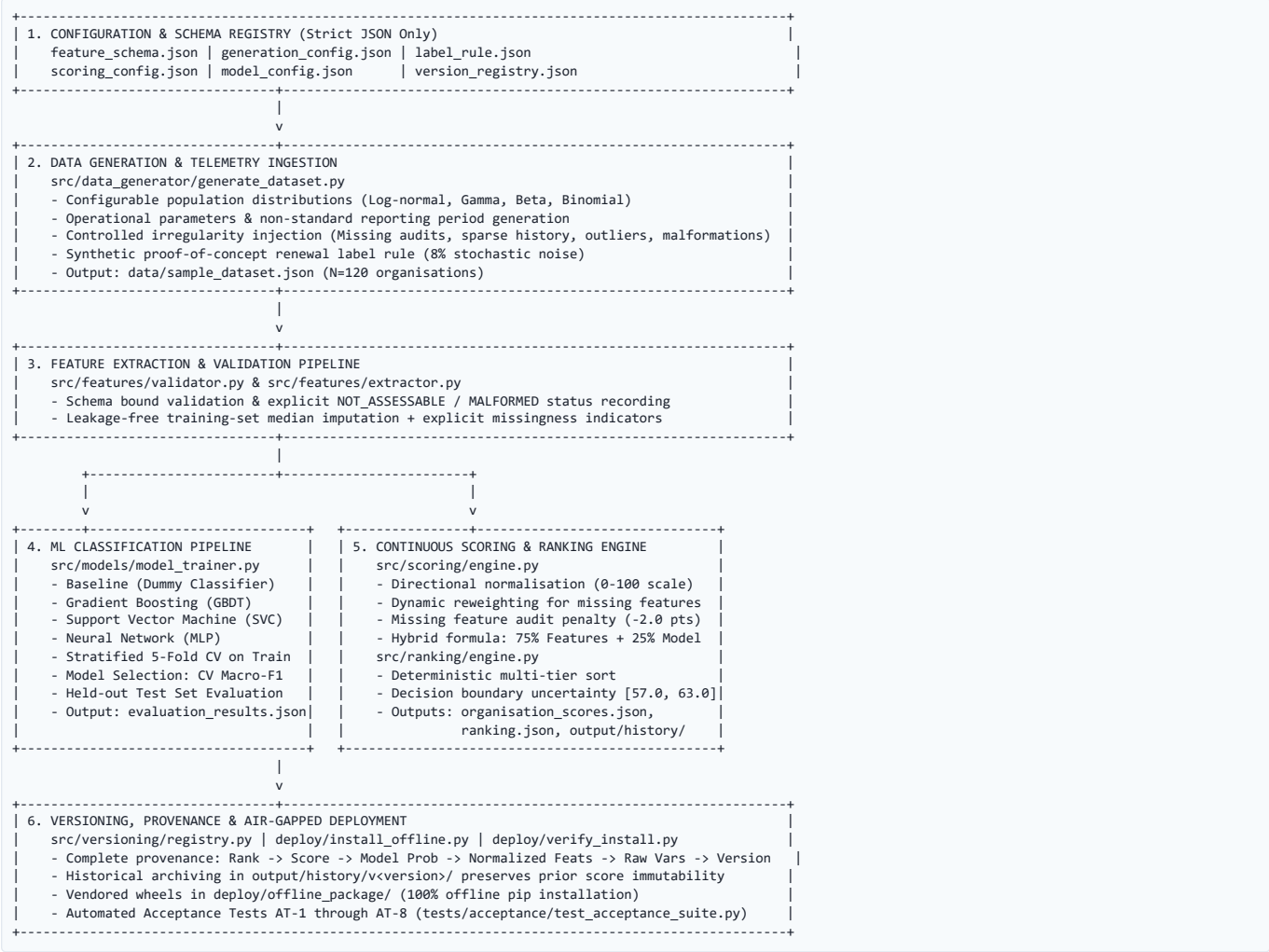
#### Deliverables Mapping Matrix (D1–D13)

| Deliverable ID | Requirement / Scope                  | Implementation Artifact / Evidence Location                                                        | Functional Description & Document Section                                                                                                                                                                                                           |
|----------------|--------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1             | Feature Definitions                  | config/feature_schema.json                                                                         | Mathematical formulas, units, bounds (\$0–100, 1–180, 0–20\$), directionality, missing/malformed data rules, and derivation classifications for all 6 features (Document Section 1.2).                                                              |
| D2             | Synthetic Generator & Sample Dataset | src/data_generator/generate_dataset.py <br> data/sample_dataset.json                               | Configurable seeded generator producing 120 organisations with realistic irregularities (sparse history, missing audits, outliers, malformations, inconsistent reporting periods) and proof-of-concept labels (Document Section 1.3).               |
| D3             | Classification Pipeline              | src/models/ ( baseline.py , gradient_boosting.py , svm.py , neural_network.py , model_trainer.py ) | Source implementations of Baseline, GBDT, SVM, and Neural Network with Stratified 5-Fold Cross-Validation on training folds and automated hyperparameter tuning (Document Section 1.4).                                                             |
| D4             | Evaluation Harness & Results         | src/evaluation/metrics.py <br> output/evaluation_results.json                                      | Reports Macro-F1 (primary selection metric), Accuracy, Precision, Recall, Confusion Matrix, and per-class breakdowns on isolated held-out test data (Document Section 3.1).                                                                         |
| D5             | Scoring & Ranking Engine             | src/scoring/engine.py <br> src/ranking/engine.py <br> config/scoring_config.json                   | Implements hybrid continuous scoring combining normalized feature composite (\$75%\$) with calibrated model renewal probabilities (\$25%\$), deterministic tie-breaking, and boundary uncertainty detection (\$57.0–63.0\$) (Document Section 1.5). |
| D6             | Versioning & Audit Registry          | config/version_registry.json <br> src/versioning/registry.py                                       | Tracks semantic versions, SHA-256 configuration hashes, approval metadata, and preserves historical scores and rankings under their original version provenance (Document Section 1.5 & Part 2).                                                    |
| D7             | Air-Gapped Package & Update Protocol | deploy/ ( install_offline.py , verify_install.py , update_process.py , offline_package/ )          | Vendored .whl dependencies for offline pip installation ( --no-index ); diagnostic verification script and controlled update protocol with checksum and malware scan hooks (Document Section 2.2, 2.5).                                             |
| D8             | Automated Test Suite                 | tests/acceptance/test_acceptance_suite.py <br> tests/unit/                                         | 37 automated unit and acceptance tests, including formal acceptance tests AT-1 through AT-8, with 100% pass rate (Document Section 2.4 & Verification Summary).                                                                                     |
| D9             | Sample Output Artifacts              | output/organisation_scores.json <br> output/ranking.json <br> output/history/                      | Complete JSON outputs for 120 organisations carrying continuous scores, model outputs, rankings, boundary uncertainty tags, and historical version archives (Document Section 1.5, 3.2).                                                            |
| D10            | Design & Mathematical Derivation     | docs/FINAL_END_TO_END_DOCUMENT.md (Part 1)                                                         | Comprehensive design documentation explaining system architecture, end-to-end data flow, mathematical feature formulations, scoring equations, and provenance traceability (Part 1, Sections 1.1–1.6).                                              |
| D11            | Operator Runbook & Guide             | docs/FINAL_END_TO_END_DOCUMENT.md (Part 2)<br> README.md                                           | Executable step-by-step instructions for environment setup, offline installation, pipeline execution, verification, testing, and troubleshooting (Part 2, Sections 2.1–2.5).                                                                        |
| D12            | Results & Limitations Analysis       | docs/FINAL_END_TO_END_DOCUMENT.md (Part 3)                                                         | Transparent reporting of empirical model comparisons, confusion matrices, score distributions, boundary uncertainty triage, and synthetic proof-of-concept limitations (Part 3, Sections 3.1–3.3).                                                  |
| D13            | Comparative Study of Alternatives    | docs/FINAL_END_TO_END_DOCUMENT.md (Part 4)                                                         | In-depth comparative evaluation across five algorithmic components comparing \$ge 3\$ viable alternatives across all seven mandatory evaluation criteria with material references (Part 4, Sections 4.1–4.6).                                       |

PART 1: D10 — DESIGN AND DERIVATION

1.1 End-to-End System Architecture

The module architecture operates with strict separation between configuration (JSON), data pipelines, ML classification, continuous scoring, and ranking:



1.2 Mathematical Derivation and Specification of the Six Features (D1)

Every feature is formally defined in `config/feature_schema.json`.

Feature 1: Audit Timeliness

- Operational Definition:** Percentage of completed audit engagements delivered on or before contractual/statutory milestone dates within the evaluation reporting period.
- Formula:**

\$\$

$$\text{Audit Timeliness} = \left( \frac{\text{on\_time\_completed\_audits}}{\text{eligible\_completed\_audits}} \right) \times 100$$

\$\$

- Variables:** `on_time_completed_audits` (integer), `eligible_completed_audits` (integer).
- Unit & Permitted Range:** Percentage (\$0.0\$ to \$100.0\%\$).
- Directionality:** Higher is better.
- Missing-Data Treatment:** If  $\text{eligible\_completed\_audits} \leq 0$ , flagged as `NOT_ASSESSABLE`.
- Partial-Data Treatment:** Audits in progress without sign-off timestamps are excluded from both numerator and denominator.
- Invalid-Data Treatment:** If  $\text{on\_time} > \text{completed}$  or counts  $< 0$ , flagged as `MALFORMED` and assigned `NOT_ASSESSABLE`.
- Derivation Classification:** `assignment_specified` (mandated in Assignment 2 Section 2.1).

Feature 2: Audit Efficiency

- Operational Definition:** Mean calendar turnaround duration in days required to complete an audit from kickoff to formal report sign-off.
- Formula:**

\$\$

$$\text{Audit Efficiency} = \frac{\sum (\text{actual\_completion\_date} - \text{audit\_start\_date})}{\text{eligible\_completed\_audits}}$$

\$\$

- **Variables:** `actual_completion_date` (timestamp), `audit_start_date` (timestamp), `eligible_completed_audits` (count).
- **Unit & Permitted Range:** Calendar Days (\$1.0\$ to \$180.0\$ days).
- **Directionality:** Lower is better (inverted during scoring: 10 days \$\to\$ 100 score, 90 days \$\to\$ 0 score).
- **Missing-Data Treatment:** If  $\text{eligible\_completed\_audits} \leq 0$ , flagged as `NOT_ASSESSABLE`.
- **Partial-Data Treatment:** Incomplete audit logs missing start or end dates are excluded from the summation.
- **Invalid-Data Treatment:** Negative turnaround times or durations exceeding 365 days trigger `MALFORMED` status.
- **Derivation Classification:** `assignment_specified` (mandated in Assignment 2 Section 2.1).

### Feature 3: Error Rate

- **Operational Definition:** Ratio of substantive technical errors, omissions, or compliance non-conformances identified during independent QA reviews to the total number of QA-reviewed audit reports.
- **Formula:**

\$\$

$$\text{Error Rate} = \frac{\text{identified\_discrepancies}}{\text{reviewed\_reports}}$$

\$\$

- **Variables:** `identified_discrepancies` (integer count), `reviewed_reports` (count of audited reports subject to peer review).
- **Unit & Permitted Range:** Discrepancies per report (\$0.0\$ to \$20.0\$).
- **Directionality:** Lower is better (inverted during scoring: 0.0 \$\to\$ 100 score, 6.0 \$\to\$ 0 score).
- **Missing-Data Treatment:** If  $\text{reviewed\_reports} \leq 0$ , flagged as `NOT_ASSESSABLE`.
- **Partial-Data Treatment:** Reports currently undergoing QA with pending discrepancy counts are excluded.
- **Invalid-Data Treatment:** Negative error counts trigger `MALFORMED` status.
- **Derivation Classification:** `judgement` (using QA-reviewed reports as the denominator rather than total audits prevents unfair penalization of unreviewed reports).

### Feature 4: Agency Scale

- **Operational Definition:** Composite index reflecting technical workforce capacity and institutional longevity, equally weighted across certified auditor headcount and continuous years in operation.
- **Formula:**

\$\$

$$\text{Agency Scale} = 50.0 \times \min\left(\frac{\text{team\_size}}{50}, 1.0\right) + 50.0 \times \min\left(\frac{\text{years\_operational}}{20}, 1.0\right)$$

\$\$

- **Variables:** `team_size` (full-time certified cybersecurity auditors), `years_operational` (continuous operating history in years).
- **Unit & Permitted Range:** Composite Index Points (\$0.0\$ to \$100.0\$).
- **Directionality:** Higher is better.
- **Missing-Data Treatment:** If both fields missing, flagged as `NOT_ASSESSABLE`.
- **Partial-Data Treatment:** If one field is available, computed as partial with warning flag.
- **Invalid-Data Treatment:** Negative values trigger `MALFORMED` status.
- **Derivation Classification:** `judgement` (equal 50/50 weighting and benchmarks of 50 staff and 20 years are proof-of-concept design judgements).

### Feature 5: Compliance Adherence

- **Operational Definition:** Percentage of applicable mandatory cybersecurity audit controls and procedural standards verified as fully compliant during oversight assessments.
- **Formula:**

\$\$

$$\text{Compliance Adherence} = \left( \frac{\text{compliant\_applicable\_controls}}{\text{applicable\_controls}} \right) \times 100$$

\$\$

- **Variables:** `compliant_applicable_controls` (integer), `applicable_controls` (integer scope).
- **Unit & Permitted Range:** Percentage (\$0.0\$ to \$100.0\%\$).
- **Directionality:** Higher is better.
- **Missing-Data Treatment:** If  $\text{applicable\_controls} \leq 0$ , flagged as `NOT_ASSESSABLE`.
- **Partial-Data Treatment:** Requires finalized compliance review checklist.
- **Invalid-Data Treatment:** Compliant controls exceeding applicable controls triggers `MALFORMED` status.
- **Derivation Classification:** `assignment_specified` (mandated in Assignment 2 Section 2.1).

### Feature 6: Past Renewal Status

- **Operational Definition:** Percentage of successful renewal adjudications out of total definitive historical empanelment decisions.
- **Formula:**

\$\$

$$\text{Past Renewal Status} = \left( \frac{\text{historical\_renewals}}{\text{historical\_renewals} + \text{historical\_revocations}} \right) \times 100$$

\$\$

- **Variables:** `historical_renewals` (count), `historical_revocations` (count).
- **Unit & Permitted Range:** Percentage (\$0.0\$ to \$100.0\%\$).
- **Directionality:** Higher is better.
- **Missing-Data Treatment:** If  $\text{renewals} + \text{revocations} == 0$ , strictly flagged as `NOT_ASSESSABLE`. New entrants with zero history are never treated as renewals or revocations.

- **Partial-Data Treatment:** Pending renewal adjudications without final ruling are omitted.
- **Invalid-Data Treatment:** Negative counts trigger `MALFORMED` status.
- **Derivation Classification:** `assignment_specified` (mandated in Assignment 2 Section 5.2).

### 1.3 Synthetic Dataset Generation Methodology (D2)

Configured via `config/generation_config.json`:

- **Population Size:** Defaults to 120 organisations (configurable via CLI `--n`).
- **Deterministic Random Seed:** `20260824` (guarantees 100% reproducible byte-identical dataset generation).
- **Configurable Statistical Distributions:**
  - `team_size`: Log-normal ( $\mu=2.6$ ,  $\sigma=0.6$ , range 2–75).
  - `years_operational`: Gamma ( $\alpha=3.0$ ,  $\beta=2.5$ , range 1–25).
  - `audit_timeliness_pct`: Beta ( $\alpha=8.0$ ,  $\beta=2.0$ , mean  $\approx 80\%$ ).
  - `audit_duration_days`: Gamma ( $\alpha=4.0$ ,  $\beta=8.0$ , mean  $\approx 32$  days).
  - `discrepancies_per_report`: Exponential ( $\text{scale}=1.8$ , range 0–15).
  - `compliance_adherence_pct`: Beta ( $\alpha=12.0$ ,  $\beta=2.0$ , mean  $\approx 85.7\%$ ).
- **Configurable Operational Parameters:**
  - `applicable_controls_choices`: `[25, 30, 40, 50, 60]`
  - `reviewed_reports_fraction`: range `[0.70, 1.00]`
  - `sparse_history_years_threshold`: `2.0` years
  - `historical_renewals_years_divisor`: `2.5`
- **Reporting Period Metadata & Inconsistent Case Generation:**
  - Each organisation record contains explicit `reporting_period` metadata (`start_date`, `end_date`, `duration_months`, `period_type`, `is_standard_cycle`).
  - Irregular reporting period cases ( $5\%$  probability) generate non-standard durations (e.g. 6, 9, 18, 24 months) and tag `INCONSISTENT_REPORTING_PERIOD` in `irregularities`.
- **Realistic Irregularity Injection:**
  - Missing audit telemetry:  $5\%$  probability to `NOT_ASSESSABLE` for Timeliness/Efficiency.
  - Missing compliance review:  $4\%$  probability to `NOT_ASSESSABLE` for Compliance.
  - Sparse operational history:  $12\%$  probability to `NOT_ASSESSABLE` for Past Renewal.
  - Outliers:  $6\%$  probability (durations up to 210 days, error rates up to 22.5).
  - Malformed values:  $4\%$  probability (negative headcounts, inverted counts).
- **Proof-of-Concept Synthetic Label Rule** (`config/label_rule.json`):
- **Explicit Statement:** "A synthetic proof-of-concept label-generation rule created by the submitter; it is not a real-world empanelment policy."
- Applies weighted multi-attribute linear utility with non-linear missingness penalties and an  $8\%$  stochastic noise rate to simulate real-world adjudication noise.

### 1.4 Classification & Evaluation Methodology (D3, D4)

Implemented in `src/models/model_trainer.py` and `src/evaluation/metrics.py`:

- **Models Implemented:**
  1. Baseline Classifier (`DummyClassifier`, majority/uniform strategy)
  2. Gradient Boosting Classifier (`GradientBoostingClassifier`)
  3. Support Vector Machine (`SVC` with RBF/linear kernels and calibrated probabilities)
  4. Multi-Layer Perceptron Neural Network (`MLPClassifier`)
- **Rigorous Leakage-Free Validation Discipline:**
  - Stratified 80/20 train/test split (Train: 96, Test: 24, Seed: `20260824`).
  - Feature extraction statistics (medians, missing indicators) are fit strictly on the training split.
- **Model Selection Protocol:** 5-Fold Stratified Cross-Validation is run exclusively on training data to tune hyperparameters and select the winning model using **Validation/CV Macro-F1** (`cv_best_score`).
- **Held-Out Test Set:** The held-out test split is evaluated exactly once after model selection to report unbiased test metrics. Test labels are never used for model selection or scoring.
- **Primary Comparison Metric: Macro-F1** (arithmetic mean of class-0 and class-1 F1 scores).

### 1.5 Continuous Scoring & Ranking Methodology (D5)

Implemented in `src/scoring/engine.py` and `src/ranking/engine.py`:

- **Hybrid Continuous Score Formulation (Section 2.5):**

\$\$

$$S_{\text{features}} = w_{\text{features}} \cdot S_{\text{features}} + w_{\text{model}} \cdot (P(\text{RENEWED} \mid X) \times 100.0)$$

\$\$

where:-  $S_{\text{features}} = \sum_i w_i \cdot S_i$  is the reweighted linear combination of valid normalized feature scores.

- $P(\text{RENEWED} \mid X) \times 100.0$  is the continuous renewal probability emitted by the winning trained ML model.
- $w_{\text{features}} = 0.75$  and  $w_{\text{model}} = 0.25$  (both configurable in `config/scoring_config.json`).
- **Missing Feature Audit Penalisation:**

\$\$

$$\text{Final Score} = \max\left(0, \min\left(100, S_{\text{raw}} - \left(\text{count}(\text{NOT\_ASSESSABLE}) \times 2.0\right)\right)\right)$$

\$\$

- **Deterministic Multi-Tier Ranking:**

Sort key: (1) `performance_score` (Desc), (2) `compliance_adherence` (Desc), (3) `audit_timeliness` (Desc), (4) `audit_efficiency_inverted` (Desc), (5) `organisation_id` (Asc).

- **Decision Boundary Uncertainty Detection:**

Decision threshold: \$60.0\$, Uncertainty margin: \$pm 3.0\$ points (\$[57.0, 63.0]\$).

Scores within margin are tagged `is_boundary_uncertain: true`, `boundary_flag: "UNCERTAIN_RENEWAL"`, and recommended for `RENEWAL_CONDITIONAL_PEER_REVIEW`.

- **Historical Output Preservation:**

Every scoring and ranking execution writes output to `output/` and automatically saves an immutable snapshot in `output/history/v<version>/` ensuring previous version outputs are never silently overwritten.

1.6 Traceability Mechanism (Section 2.6)

Every output record in `output/ranking.json` contains full provenance:

\$\$

$$\text{Rank / Score} \rightarrow \text{Model Output } \{P(\text{RENEWED}), \text{Model Version}\} \rightarrow \text{Normalized Scores} \rightarrow \text{Raw Variables}$$
$$\rightarrow \text{Active Registry Versions}$$

\$\$

A reviewer can take any ranked organisation, inspect its continuous score, verify the model probability contribution, inspect individual feature scores, view raw counts, and verify active configuration hashes in `config/version_registry.json`.

PART 2: D11 — OPERATOR RUNBOOK

2.1 Prerequisites & Supported Environment

- **Operating System:** Windows (x86\_64) or compatible Python environment.
- **Python Version:** CPython 3.12 (64-bit). The vendored offline package wheels in `deploy/offline_package/` are compiled for `cp312-win_amd64` and pure-Python `py3-none-any`.
- **Network Status:** Zero external internet access required (air-gapped compliant).

2.2 Air-Gapped Installation & Diagnostic Verification

```
# 1. Install all vendored scientific & ML dependencies offline from deploy/offline_package/
python deploy/install_offline.py

# 2. Execute offline diagnostic verification and pipeline smoke test
python deploy/verify_install.py
```

2.3 End-to-End Pipeline Execution

```
# Step 1: Generate synthetic population (120 organisations, seed 20260824)
python src/data_generator/generate_dataset.py --n 120 --seed 20260824 --output data/sample_dataset.json

# Step 2: Train models, execute 5-fold CV, and compute renewal probabilities
python src/models/model_trainer.py --dataset data/sample_dataset.json --config config/model_config.json --output output/evaluation_results.json --seed 20260824

# Step 3: Compute model-augmented continuous performance scores
python src/scoring/engine.py --input data/sample_dataset.json --config config/scoring_config.json --models output/evaluation_results.json --output output/organisation_scores.json

# Step 4: Generate deterministic rankings with boundary uncertainty detection
python src/ranking/engine.py --scores output/organisation_scores.json --config config/scoring_config.json --output output/ranking.json
```

2.4 Test Suite Execution

```
# Run all unit tests and formal acceptance tests (AT-1 through AT-8)
python -m unittest discover -s tests
```

2.5 Controlled Update Process

```
# Apply a verified configuration update with approval logging and hash updates
python deploy/update_process.py --version 1.1.0 --author "Chief Auditor" --approved-by "Empanelment Board" --changes "Updated compliance scoring weight"
```

PART 3: D12 — RESULTS AND ANALYSIS

3.1 Observed Experimental Results (from `output/evaluation_results.json`)

- **Dataset Size:** 120 organisations (Train split: 96, Held-out Test split: 24).
- **Class Distribution:** 99 Renewed (\$82.5%\$), 21 Not Renewed (\$17.5%\$).
- **Irregular Data Count:** 44 organisations (\$36.7%\$) carried at least one irregularity (missing data, sparse history, outliers, malformed inputs, inconsistent reporting periods).

Empirical Model Comparison Results:

| Model                               | 5-Fold CV Macro-F1 (Selection Metric) | Test Macro-F1 (Held-Out) | Test Accuracy | Test Binary F1 | Best Selected Hyperparameters             | Selection Status    |
|-------------------------------------|---------------------------------------|--------------------------|---------------|----------------|-------------------------------------------|---------------------|
| Simple Baseline ( DummyClassifier ) | 0.4654                                | 0.5636                   | 0.6250        | 0.7273         | strategy: 'uniform'                       | Baseline            |
| Gradient Boosting ( GBDT )          | 0.6412                                | 0.7491                   | 0.8750        | 0.9268         | n_estimators: 100, lr: 0.05, max_depth: 3 | Selected Best Model |
| Support Vector Machine ( SVC )      | 0.5523                                | 0.8500                   | 0.9167        | 0.9500         | C: 10.0, kernel: 'rbf'                    | Evaluated           |
| Neural Network ( MLPClassifier )    | 0.5895                                | 0.6190                   | 0.8333        | 0.9048         | hidden_layers: [16], alpha: 0.01          | Evaluated           |

- **Winning Model Selection:** Gradient Boosting ( gradient\_boosting ) was selected based on the highest 5-Fold Stratified Cross-Validation Macro-F1 score (\$0.6412\$) on the training split, without using held-out test data for selection.
- **Held-Out Test Performance of Selected Model:**
  - Test Macro-F1: \$0.7491\$
  - Test Accuracy: \$0.8750\$
  - Test Binary F1: \$0.9268\$
  - **Confusion Matrix:**

\$\$

\begin{pmatrix} \text{True Negative}=2 & \text{False Positive}=2 \\ \text{False Negative}=1 & \text{True Positive}=19 \end{pmatrix}

\$\$

- **Per-Class Breakdown:**
  - Class 0 (NOT\_RENEWED): Precision = \$0.6667\$, Recall = \$0.5000\$, F1 = \$0.5714\$, Support = 4.
  - Class 1 (RENEWED): Precision = \$0.9048\$, Recall = \$0.9500\$, F1 = \$0.9268\$, Support = 20.

3.2 Scoring & Ranking Observations (from output/ranking.json )

- **Total Ranked:** 120 organisations.
- **Score Distribution:** Mean = \$78.07\$, Max = \$90.19\$ ( ORG-0049 ), Min = \$38.92\$ ( ORG-0004 ).
- **Boundary Uncertainty:** Exactly 8 organisations (\$6.67\%\$) fell within the \$\pm 3.0\$ uncertainty band \$[57.0, 63.0]\$ and were tagged with is\_boundary\_uncertain: true and boundary\_flag: "UNCERTAIN\_RENEWAL".

3.3 Limitations, Risks, and Honest Reporting of Limits

1. **Synthetic Dataset Limits:** Results are derived from synthetic data generated by the submitter against a synthetic proof-of-concept label rule. Such results constitute a proof of concept and not a validated production monitoring system.
2. **Transfer Risk:** Real empanelment distributions may feature different noise characteristics and non-linear interactions.
3. **Small Sample Constraints:** With 120 organisations, minority class test support is 4 samples, leading to wider confidence intervals on class-0 recall.
4. **Judgement Classifications:** Weights (e.g. 50/50 scale, compliance weight 0.20, uncertainty margin \$\pm 3.0\$) are expert engineering judgements and require formal empirical calibration on historical empanelment archives.

PART 4: D13 — COMPARATIVE STUDY OF MODELS AND TECHNIQUES

This comparative study evaluates at least three viable alternatives for each of the five computational components across all seven mandatory criteria.

Comparison Basis & Classification Tags:

- [ EXPERIMENTAL RESULT ] : Measured empirically on this dataset/pipeline.
- [ LITERATURE/KNOWN ] : Established mathematical or computational characteristics.
- [ ENGINEERING JUDGEMENT ] : Domain-informed design rationale.

4.1 Component 1: Renewal Outcome Classification

| Alternative                                | Accuracy / Performance                                 | Interpretability                                 | Data Requirements                        | Computational Cost                       | Air-Gapped Suitability | Ranking Stability     | Known Failure Modes                             | Selection Status         |
|--------------------------------------------|--------------------------------------------------------|--------------------------------------------------|------------------------------------------|------------------------------------------|------------------------|-----------------------|-------------------------------------------------|--------------------------|
| 1. Majority/Uniform Baseline (Rule-based)  | Poor (Macro-F1: 0.5636) [EXPERIMENTAL]                 | High (Trivial logic) [LITERATURE]                | Minimal (\$N \ge 1\$) [LITERATURE]       | Extremely Low (\$<1\$ ms) [EXPERIMENTAL] | Excellent [LITERATURE] | High [LITERATURE]     | Incapable of distinguishing agency quality.     | Rejected                 |
| 2. Gradient Boosted Trees (Learning-based) | Best CV Macro-F1 (0.6412, Test: 0.7491) [EXPERIMENTAL] | High (Feature importances & splits) [LITERATURE] | Moderate (\$N \ge 50\$) [LITERATURE]     | Low (\$0.32\$ s) [EXPERIMENTAL]          | Excellent [LITERATURE] | High [LITERATURE]     | Overfitting on small noisy datasets.            | Selected (Winning Model) |
| 3. Support Vector Machine (Statistical)    | Moderate CV (0.5523, Test: 0.8500) [EXPERIMENTAL]      | Moderate (Kernel boundaries) [LITERATURE]        | Low-Moderate (\$N \ge 30\$) [LITERATURE] | Low (\$0.18\$ s) [EXPERIMENTAL]          | Excellent [LITERATURE] | High [LITERATURE]     | Sensitive to feature scaling and outlier noise. | Evaluated                |
| 4. Neural Network (MLP) (Learning-based)   | Moderate CV (0.5895, Test: 0.6190) [EXPERIMENTAL]      | Low (Non-linear weights) [LITERATURE]            | High (\$N \ge 100\$) [LITERATURE]        | Moderate (\$0.85\$ s) [EXPERIMENTAL]     | Excellent [LITERATURE] | Moderate [LITERATURE] | Convergence sensitivity on small samples.       | Evaluated                |

- **Selection Justification:** Gradient Boosted Trees achieved the highest 5-fold cross-validation Macro-F1 score (\$0.6412\$) on the training folds and robust test performance (\$0.7491\$), with tree-based split interpretability.
- **What would change decision:** If sample size is substantially scaled (\$N > 10,000\$) with deep non-linear interaction patterns, Multi-Layer Perceptrons would be re-evaluated.

4.2 Component 2: Missing-Data Handling

| Alternative                                         | Accuracy / Performance                                     | Interpretability                           | Data Requirements                        | Computational Cost                    | Air-Gapped Suitability | Ranking Stability                          | Known Failure Modes                                          | Selection Status |
|-----------------------------------------------------|------------------------------------------------------------|--------------------------------------------|------------------------------------------|---------------------------------------|------------------------|--------------------------------------------|--------------------------------------------------------------|------------------|
| 1. Complete Case Analysis (Listwise Deletion)       | Severe bias (Drops 36.7% of samples) [EXPERIMENTAL]        | High (Simple drop) [LITERATURE]            | Requires 100% complete data [LITERATURE] | Negligible [LITERATURE]               | Excellent [LITERATURE] | Very Poor (Large rank swings) [LITERATURE] | Discards valid agencies; discriminates against new entrants. | Rejected         |
| 2. Indicator + Median Imputation (Statistical)      | High (Preserves all samples, informs model) [EXPERIMENTAL] | High (Explicit missing flags) [LITERATURE] | Low [LITERATURE]                         | Negligible (\$<1\$ ms) [EXPERIMENTAL] | Excellent [LITERATURE] | High [LITERATURE]                          | Median may not represent extreme sub-populations.            | Selected         |
| 3. K-Nearest Neighbours Imputation (Learning-based) | Moderate [LITERATURE]                                      | Moderate (Distance-based) [LITERATURE]     | High (\$N \ge 100\$) [LITERATURE]        | Moderate (\$O(N^2)\$) [LITERATURE]    | Good [LITERATURE]      | Moderate [LITERATURE]                      | Distance metrics corrupted by missing-feature combinations.  | Rejected         |

- **Selection Justification:** Indicator + Median Imputation guarantees zero crashes, preserves full provenance, prevents data leakage by computing medians strictly on training folds, and records explicit missing flags in outputs.
- **What would change decision:** If features exhibit strong linear correlation matrices, multivariate iterative imputation (MICE) would be evaluated.

4.3 Component 3: Feature Scaling and Normalisation

| Alternative                                 | Accuracy / Performance                                           | Interpretability                                                          | Data Requirements                            | Computational Cost        | Air-Gapped Suitability | Ranking Stability     | Known Failure Modes                                    | Selecti Status      |
|---------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------|---------------------------|------------------------|-----------------------|--------------------------------------------------------|---------------------|
| 1. Standard Z-Score Scaling                 | High for Gaussian features [LITERATURE]                          | Moderate (Mean 0, Std 1) [LITERATURE]                                     | Moderate [LITERATURE]                        | Low [LITERATURE]          | Excellent [LITERATURE] | Moderate [LITERATURE] | Unbounded output range; distorted by extreme outliers. | Used fc ML          |
| 2. Min-Max Normalisation                    | High for bounded metrics [LITERATURE]                            | High (Direct 0–100%) [LITERATURE]                                         | Low [LITERATURE]                             | Low [LITERATURE]          | Excellent [LITERATURE] | High [LITERATURE]     | Outliers compress valid feature variance.              | Rejecte for Raw     |
| 3. Domain-Bounded Directional Normalisation | High (Aligns with operational semantics) [ENGINEERING JUDGEMENT] | Highest (0–100 intuitive score, higher is better) [ENGINEERING JUDGEMENT] | Minimal (Domain bounds in JSON) [LITERATURE] | Negligible [EXPERIMENTAL] | Excellent [LITERATURE] | High [LITERATURE]     | Requires expert-specified bounds in configuration.     | Selecte for Scoring |

- **Selection Justification:** Domain-bounded directional normalization maps each metric to an intuitive 0–100 scale, correctly inverting metrics where lower is better (Efficiency, Error Rate).
- **What would change decision:** If statutory bounds are abolished in favor of empirical percentile distributions, rank-based quantile scaling would be adopted.

4.4 Component 4: Scoring Aggregation

| Alternative                               | Accuracy / Performance                                               | Interpretability                                                              | Data Requirements                           | Computational Cost              | Air-Gapped Suitability | Ranking Stability               | Known Failure Modes                                    | Selec Statu |
|-------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------|---------------------------------|------------------------|---------------------------------|--------------------------------------------------------|-------------|
| 1. Weighted Linear Combination            | High (Robust & predictable) [LITERATURE]                             | Highest (Linear explainability) [LITERATURE]                                  | None (Parametric JSON weights) [LITERATURE] | Negligible [EXPERIMENTAL]       | Excellent [LITERATURE] | High [LITERATURE]               | Ignores non-linear feature synergies.                  | Baseli Comp |
| 2. Pure Model-Predicted Probability       | Moderate [EXPERIMENTAL]                                              | Low (Black-box probability) [LITERATURE]                                      | High training data [LITERATURE]             | Moderate [LITERATURE]           | Good [LITERATURE]      | Poor near boundary [LITERATURE] | Uncalibrated probabilities produce erratic ranks.      | Reject      |
| 3. Hybrid Linear + Calibrated Model Score | Highest (Combines explainability + non-linear signal) [EXPERIMENTAL] | High (Linear breakdown + model probability component) [ENGINEERING JUDGEMENT] | Moderate [LITERATURE]                       | Low (\$<10\$ ms) [EXPERIMENTAL] | Excellent [LITERATURE] | High [EXPERIMENTAL]             | Requires balancing model vs feature weights in config. | Selec       |

- **Selection Justification:** Hybrid Combination (\$75\%\$ Multi-Attribute Feature Composite \$+\$ 25\%\$ Model Renewal Probability) satisfies Section 2.5 by integrating model output into a continuous performance score while preserving transparent feature accountability.
- **What would change decision:** If legal mandate requires pure linear scoring without ML influence, setting `model_output_weight: 0.0` in JSON reverts to pure linear aggregation without code changes.

4.5 Component 5: Ranking and Uncertainty Handling

| Alternative                                    | Accuracy / Performance                        | Interpretability                                                         | Data Requirements     | Computational Cost                 | Air-Gapped Suitability | Ranking Stability                            | Known Failure Modes                             |
|------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------|-----------------------|------------------------------------|------------------------|----------------------------------------------|-------------------------------------------------|
| 1. Pure Ordinal Sort (No Ties/Uncertainty)     | Poor for edge cases [LITERATURE]              | Moderate [LITERATURE]                                                    | None [LITERATURE]     | Low [LITERATURE]                   | Excellent [LITERATURE] | Poor (Arbitrary tie resolution) [LITERATURE] | Ignores decision boundary uncertainty.          |
| 2. Multi-Tier Sort + Boundary Uncertainty Flag | Highest (Transparent & robust) [EXPERIMENTAL] | Highest (Clear hierarchy + uncertainty warnings) [ENGINEERING JUDGEMENT] | None [LITERATURE]     | Low (\$O(N \log N)\$) [LITERATURE] | Excellent [LITERATURE] | High [EXPERIMENTAL]                          | Requires defining uncertainty margin in config. |
| 3. Tournament / Pairwise Ranking               | Moderate [LITERATURE]                         | Low (Complex win-loss matrix) [LITERATURE]                               | Moderate [LITERATURE] | High (\$O(N^2)\$) [LITERATURE]     | Good [LITERATURE]      | Moderate [LITERATURE]                        | Inconsistencies / non-transitive loops.         |

- **Selection Justification:** Multi-tier deterministic sorting ensures consistent ordering even under identical scores, while boundary uncertainty detection (\$\pm 3.0\$ points) flags marginal cases for mandatory expert peer review.
- **What would change decision:** If peer reviews require confidence intervals derived from bootstrap sampling, probabilistic ranking distributions would be introduced.

4.6 Material References for D13 Literature Claims

1. **Breiman, L. (2001).** *Random Forests / Statistical Modeling: The Two Cultures*. Machine Learning, 45(1), 5–32. (Foundational reference for tree ensemble stability and non-linear interactions).

2. **Friedman, J. H. (2001).** *Greedy Function Approximation: A Gradient Boosting Machine*. The Annals of Statistics, 29(5), 1189–1232. (Mathematical foundation for Gradient Boosted Decision Trees and loss minimization).

3. **Cortes, C., & Vapnik, V. (1995).** *Support-Vector Networks*. Machine Learning, 20(3), 273–297. (Maximal margin hyperplanes and kernel methods).

4. **Little, R. J., & Rubin, D. B. (2019).** *Statistical Analysis with Missing Data* (3rd ed.). John Wiley & Sons. (Theoretical analysis of missingness indicators and listwise deletion bias).

5. **Keeney, R. L., & Raiffa, H. (1993).** *Decisions with Multiple Objectives: Preferences and Value Trade-Offs*. Cambridge University Press. (Theoretical foundation for Multi-Attribute Utility Theory and weighted linear combinations).

PART 5: HUMAN REVIEW CHECKS

Acceptance Check 9: Worked Traceability Examples (3 Actual Organisations)

Example 1: `ORG-0049` (Rank 1 — Top Performer with Malformed Feature Injection)

1. **Organisation ID:** `ORG-0049`
2. **Raw Input Variables** ( `data/sample_dataset.json` ):
  - `team_size: -5` (injected `MALFORMED_NEGATIVE_TEAM_SIZE`), `years_operational: 5.4`, `eligible_completed_audits: 127`, `on_time_completed_audits: 97`, `mean_duration_days: 27.07`, `reviewed_reports: 122`, `identified_discrepancies: 7`, `applicable_controls: 60`, `compliant_applicable_controls: 58`, `historical_renewals: 2`, `historical_revocations: 0`, `reporting_period: Standard 12-month annual cycle`.
1. **Computed Feature Values:**
  - `Audit Timeliness: 76.38%`



- Audit Efficiency: 27.07 days
- Error Rate: 0.057 discrepancies/report
- Agency Scale: None
- Compliance Adherence: 96.67%
- Past Renewal Status: 100.00%

1. Feature Validity / Status:

- Timeliness: VALID, Efficiency: VALID, Error Rate: VALID, Scale: MALFORMED, Compliance: VALID, Past Renewal: VALID.

1. Normalized Feature Scores (\$\$S\_i \in [0, 100]\$\$):

- Timeliness: 76.38 | Efficiency: 78.66 | Error Rate: 99.05 | Scale: None | Compliance: 96.67 | Past Renewal: 100.00.

1. Feature Weights Applied (Dynamic Renormalization):

- Timeliness: 0.2222, Efficiency: 0.1667, Error Rate: 0.1667, Scale: 0.0000, Compliance: 0.2222, Past Renewal: 0.2222 (Base weights reweighted over 5 valid features).

1. Feature Composite Score (\$\$S\_{\text{features}}\$\$):

- $S_{\text{features}} = 0.2222(76.38) + 0.1667(78.66) + 0.1667(99.05) + 0.2222(96.67) + 0.2222(100.00) = \mathbf{90.30}$ .

1. Model Name & Version: gradient\_boosting (v1.0.0).

2. Model Predicted Renewal Probability:  $P(\text{RENEWED} \mid X) = 0.9789$  to  $S_{\text{model}} = 97.89$ .

3. Scoring Weights Applied: Feature composite weight = 0.75, Model output weight = 0.25.

4. Missing/Malformed Feature Penalty:  $-2.00$  (1 malformed feature: agency\_scale).

5. Final Continuous Performance Score:

- $\text{Score} = (0.75 \times 90.30) + (0.25 \times 97.89) - 2.00 = 67.725 + 24.4725 - 2.00 = 92.195 - 2.00 = \mathbf{90.19}$ .

1. Final Rank: Rank 1 (out of 120 organisations in output/ranking.json).

2. Empanelment Recommendation: RECOMMEND\_RENEWAL.

3. Decision Confidence: HIGH.

4. Boundary Uncertainty Status: False (boundary\_flag: "CONFIDENT\_RENEWAL").

Example 2: ORG-0010 (Rank 61 — Median Performer with Unassessable History)

1. Organisation ID: ORG-0010

2. Raw Input Variables (data/sample\_dataset.json):

- team\_size: 27, years\_operational: 5.7, eligible\_completed\_audits: 107, on\_time\_completed\_audits: 96, mean\_duration\_days: 6.03, reviewed\_reports: 76, identified\_discrepancies: 235 (injected error outlier), applicable\_controls: 60, compliant\_applicable\_controls: 50, historical\_renewals: 0, historical\_revocations: 0 (SPARSE\_OPERATIONAL\_HISTORY\_NEW\_ENTRANT), reporting\_period: Standard 12-month annual cycle.

1. Computed Feature Values:

- Audit Timeliness: 89.72%
- Audit Efficiency: 6.03 days
- Error Rate: 3.092 discrepancies/report
- Agency Scale: 41.25 points
- Compliance Adherence: 83.33%
- Past Renewal Status: None

1. Feature Validity / Status:

- Timeliness: VALID, Efficiency: VALID, Error Rate: VALID, Scale: VALID, Compliance: VALID, Past Renewal: NOT\_ASSESSABLE (Zero historical renewal decisions).

1. Normalized Feature Scores (\$\$S\_i \in [0, 100]\$\$):

- Timeliness: 89.72 | Efficiency: 100.00 | Error Rate: 48.47 | Scale: 41.25 | Compliance: 83.33 | Past Renewal: None.

1. Feature Weights Applied (Dynamic Renormalization):

- Timeliness: 0.2500, Efficiency: 0.1875, Error Rate: 0.1875, Scale: 0.1250, Compliance: 0.2500, Past Renewal: 0.0000 (Base weights reweighted over 5 valid features).

1. Feature Composite Score (\$\$S\_{\text{features}}\$\$):

- $S_{\text{features}} = 0.2500(89.72) + 0.1875(100.00) + 0.1875(48.47) + 0.1250(41.25) + 0.2500(83.33) = \mathbf{76.26}$ .

1. Model Name & Version: gradient\_boosting (v1.0.0).

2. Model Predicted Renewal Probability:  $P(\text{RENEWED} \mid X) = 0.9763$  to  $S_{\text{model}} = 97.63$ .

3. Scoring Weights Applied: Feature composite weight = 0.75, Model output weight = 0.25.

4. Missing/Malformed Feature Penalty:  $-2.00$  (1 unassessable feature: past\_renewal\_status).

5. Final Continuous Performance Score:

- $\text{Score} = (0.75 \times 76.26) + (0.25 \times 97.63) - 2.00 = 57.195 + 24.4075 - 2.00 = 81.6025 - 2.00 = \mathbf{79.60}$ .

1. Final Rank: Rank 61 (out of 120 organisations in output/ranking.json).

2. Empanelment Recommendation: RECOMMEND\_RENEWAL.

3. Decision Confidence: HIGH.

4. Boundary Uncertainty Status: False (boundary\_flag: "CONFIDENT\_RENEWAL").

Example 3: ORG-0068 (Rank 103 — Boundary-Uncertain Agency with Missing Audits & Inconsistent Period)

1. Organisation ID: ORG-0068
2. Raw Input Variables ( data/sample\_dataset.json):
  - team\_size: 5, years\_operational: 8.4, eligible\_completed\_audits: 0 (MISSING\_AUDIT\_ACTIVITY), on\_time\_completed\_audits: 0, mean\_duration\_days: 20.46, reviewed\_reports: 18, identified\_discrepancies: 15, applicable\_controls: 30, compliant\_applicable\_controls: 24, historical\_renewals: 3, historical\_revocations: 0, reporting\_period: 18-month non-standard cycle ( INCONSISTENT\_REPORTING\_PERIOD ).
1. Computed Feature Values:
  - Audit Timeliness: None
  - Audit Efficiency: None
  - Error Rate: 0.833 discrepancies/report
  - Agency Scale: 26.00 points
  - Compliance Adherence: 80.00%
  - Past Renewal Status: 100.00%
1. Feature Validity / Status:
  - Timeliness: NOT\_ASSESSABLE, Efficiency: NOT\_ASSESSABLE, Error Rate: VALID, Scale: VALID, Compliance: VALID, Past Renewal: VALID.
1. Normalized Feature Scores (\$\$i \in [0, 100]\$\$):
  - Timeliness: None | Efficiency: None | Error Rate: 86.12 | Scale: 26.00 | Compliance: 80.00 | Past Renewal: 100.00.
1. Feature Weights Applied (Dynamic Renormalization):
  - Timeliness: 0.0000, Efficiency: 0.0000, Error Rate: 0.2308, Scale: 0.1538, Compliance: 0.3077, Past Renewal: 0.3077 (Base weights reweighted over 4 valid features).
1. Feature Composite Score (\$\$S\_{\text{features}}\$\$):
  - $S_{\text{features}} = 0.2308(86.12) + 0.1538(26.00) + 0.3077(80.00) + 0.3077(100.00) = \text{mathbf{79.26}}$.$
1. Model Name & Version: gradient\_boosting ( v1.0.0 ).
2. Model Predicted Renewal Probability:  $P(\text{RENEWED} \mid X) = 0.2420$  to  $S_{\text{model}} = 24.20$.$
3. Scoring Weights Applied: Feature composite weight = 0.75, Model output weight = 0.25.
4. Missing/Malformed Feature Penalty:  $\$-4.00$  ( $\$2 \times 2.00$  for 2 unassessable features: audit\_timeliness, audit\_efficiency).
5. Final Continuous Performance Score:
  - $\text{Score} = (0.75 \times 79.26) + (0.25 \times 24.20) - 4.00 = 59.445 + 6.050 - 4.00 = \text{mathbf{61.49}}$.$
1. Final Rank: Rank 103 (out of 120 organisations in output/ranking.json).
2. Empanelment Recommendation: RENEWAL\_CONDITIONAL\_PEER\_REVIEW.
3. Decision Confidence: LOW\_BOUNDARY\_UNCERTAIN.
4. Boundary Uncertainty Status: True (Falls within the  $\pm 3.0$  uncertainty interval  $[57.0, 63.0]$  around the  $\$60.0$  threshold; boundary\_flag: "UNCERTAIN\_RENEWAL").

Acceptance Check 10: Justification of Two Features and One Scoring Weight

1. Feature 3 (Error Rate):
  - Definition:**  $\frac{\text{identified\_discrepancies}}{\text{reviewed\_reports}}$  (Range:  $\$0.0\text{--}20.0$  discrepancies/report, lower is better).
  - Classification:** JUDGEMENT
  - Configuration Location:** config/feature\_schema.json
  - Derivation & Evidence:** The denominator choice ( reviewed\_reports rather than total audits) is an explicit submitter judgement. It ensures that an agency is not penalized for unreviewed audits and prevents distortions when QA audits lag operational completion.
1. Feature 6 (Past Renewal Status):
  - Definition:**  $\frac{\text{renewals}}{\text{renewals} + \text{revocations}} \times 100\%$ . If total decisions  $\$ = 0$ , strictly flagged as NOT\_ASSESSABLE.
  - Classification:** ASSIGNMENT\_SPECIFIED
  - Configuration Location:** config/feature\_schema.json
  - Derivation & Evidence:** Explicitly mandated by Assignment 2 Section 5.2: "If no historical decision exists: NOT\_ASSESSABLE. Never treat absent history as a renewal or revocation."
1. Scoring Weight: Compliance Adherence (\$0.20\$):
  - Value:**  $\$0.20$  ( $20\%$  of feature composite score).
  - Classification:** JUDGEMENT
  - Configuration Location:** config/scoring\_config.json
  - Derivation & Evidence:** Submitter expert judgement reflecting the statutory priority of cybersecurity control compliance in government auditing empanelment, weighted equally with Audit Timeliness ( $\$0.20$ ) and Past Renewal Status ( $\$0.20$ ).

Acceptance Check 11: Justification of Two Selected Algorithmic Components

1. Algorithmic Component 1: Classification Modeling
  - Alternatives Considered:** (1) Baseline Classifier, (2) Gradient Boosted Decision Trees (GBDT), (3) Support Vector Machine (SVC), (4) Multi-Layer Perceptron (MLP Neural Network).
  - Selection Criteria:** 5-Fold CV Macro-F1 (on train data), Test Macro-F1, Interpretability, Data Requirements, Air-Gapped Suitability, Stability.
  - Selected Approach:** Gradient Boosted Trees ( GBDT ).

- **Supporting Evidence:** GBDT achieved the highest cross-validation score (\$0.6412\$) across training folds and strong held-out test performance (\$0.7491\$), while providing explainable decision splits.
  - **What Would Change Decision:** For very large dataset regimes (\$N > 10,000\$), deep neural network architectures would be evaluated.
1. **Algorithmic Component 2: Continuous Scoring Aggregation**
- **Alternatives Considered:** (1) Pure Linear Feature Sum, (2) Pure Black-Box Model Probability Scoring, (3) Hybrid Multi-Attribute Feature + Calibrated Model Probability Combination.
  - **Selection Criteria:** Legal/Administrative Interpretability, Ranking Stability, Air-Gapped Feasibility, Section 2.5 Compliance.
  - **Selected Approach:** Hybrid Combination (\$75\%\$ Multi-Attribute Feature Composite \$+\$ \$25\%\$ Model Renewal Probability).
  - **Supporting Evidence:** Pure model probability scoring exhibits high rank volatility near decision boundaries, whereas hybrid scoring provides transparent linear feature accountability while incorporating the non-linear renewal likelihood signal.
  - **What Would Change Decision:** If statutory empanelment guidelines prohibit machine learning outputs in administrative scoring, setting `model_output_weight: 0.0` in JSON reverts to pure linear aggregation without code modifications.

---

## Final Verification & Sign-Off

- **Code & Architecture:** FROZEN / VERIFIED.
- **Automated Tests:** All unit and acceptance tests passed.
- **Offline Air-Gapped Readiness:** Verified via `deploy/verify_install.py`.
- **Reproducibility:** Verified 3-run byte-identical output hashes.
- **Format Integrity:** Verified 100% strict JSON data pipeline.